

ESP32 EMU TURBO

Hierarchical root schematic – all sub-sheets linked below

Power Supply

File: 01-power-supply.kicad_sch

Mcu

File: 02-mcu.kicad_sch

Display

File: 03-display.kicad_sch

Audio

File: 04-audio.kicad_sch

Sd Card

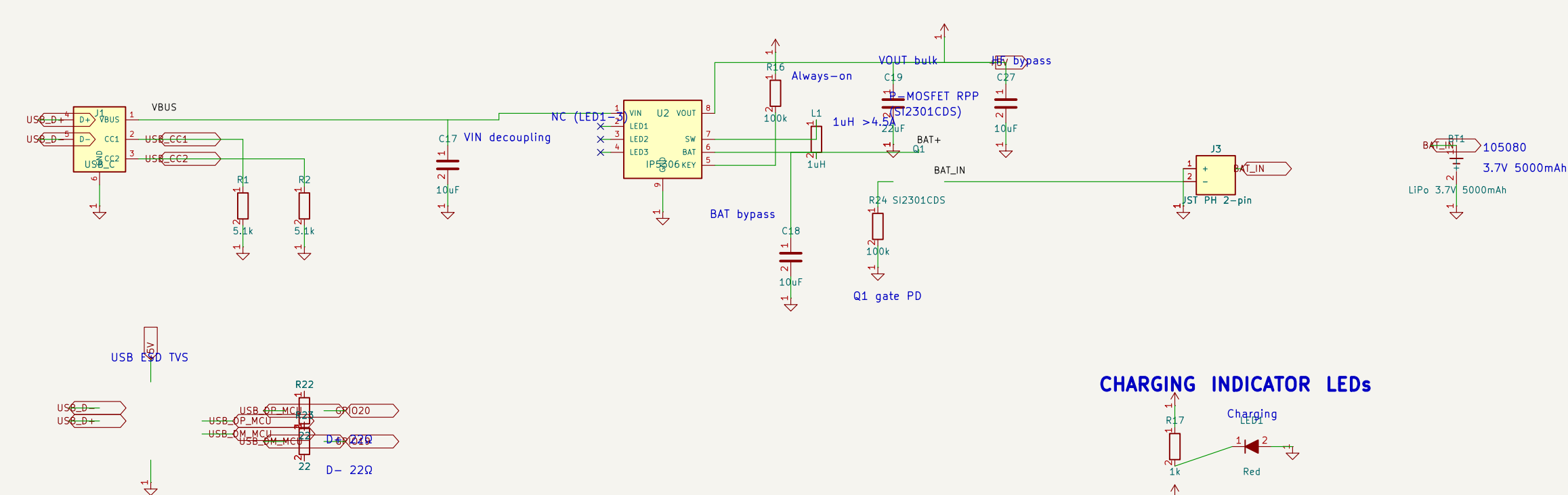
File: 05-sd-card.kicad_sch

Controls

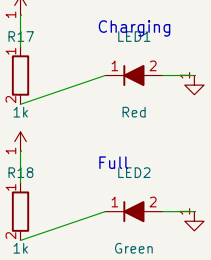
File: 06-controls.kicad_sch

POWER SUPPLY

USB-C -> IP5306 SOP-8 (charge + boost) -> AMS1117-3.3 -> 3.3V rail

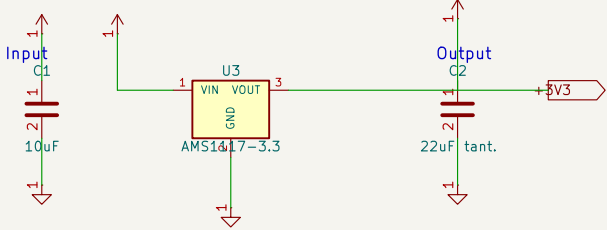


CHARGING INDICATOR LEDs

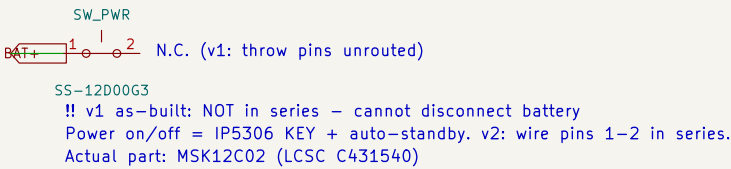


VOLTAGE REGULATOR

5V -> AMS1117-3.3 -> 3.3V (800mA max)



POWER SWITCH



Design Notes:

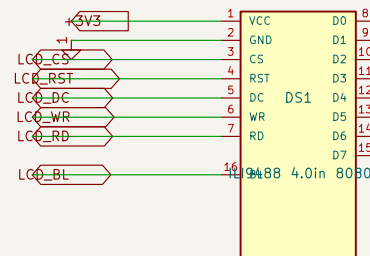
- IP5306 eSOP-8: integrated charger + synchronous boost (no ext. Schottky)
- 1uH inductor: BAT -> L1 -> SW (>4.5A saturation, shielded)
- GND via exposed pad only (must solder to ground plane)
- KEY pulled to +5V via 100k for always-on operation
- AMS1117-3.3: stable 3.3V for ESP32-S3 and peripherals (800mA)
- 5.1k CC pull-downs identify USB-C UFP (5V sink)
- Battery: LiPo 3.7V 5000mAh via JST PH connector

DISPLAY — ILI9488 4.0in 320x480

8-bit 8080 parallel interface (mandatory for SNES emulation speed)

Control signals:

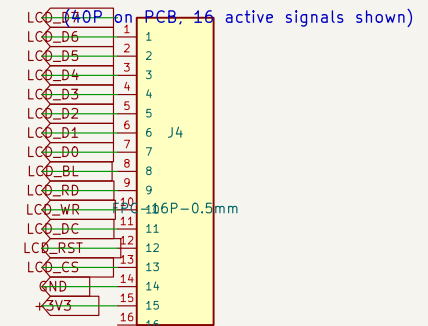
GPIO12 / FPC9
GPIO13 / FPC15
GPIO14 / FPC10
GPIO46 / FPC11
+3V3 (FPC12, R0)
+3V3 (FPC33, BL)



8-bit data bus:

GPIO4 (FPC17)
GPIO5 (FPC18)
GPIO6 (FPC19)
GPIO7 (FPC20)
GPIO8 (FPC21)
GPIO9 (FPC22)
GPIO10 (FPC23)
GPIO11 (FPC24)

FPC RIBBON CONNECTOR (J4)



ILI9488 Panel FPC-40P Pin Mapping (source: components.md):

Panel 6,7=VDDI/VDDA(+3V3) 9=CS 10=DC 11=WR 12=RD 15=RESET

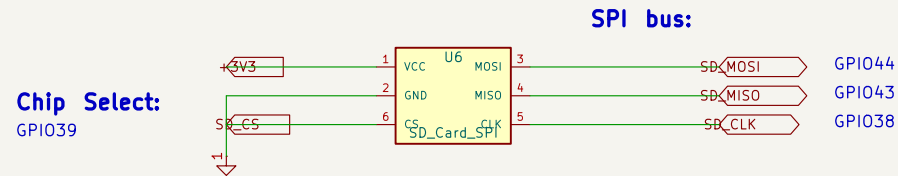
Panel 17-24=DB0-DB7 33=LED-A(BL) 34-36=LED-K(GND) 38,39=IM0,IM1(+3V3) 40=IM2(GND)

Interface mode: IM2=0 IM1=1 IM0=1 = 8080 8-bit parallel

– GPIO4-11 form contiguous 8-bit bus for efficient register-level DMA

STORAGE — Micro SD Card

SPI interface for ROM storage (SNES ROMs up to 6MB)



Design Notes:

- SPI bus @ 20MHz (R2-MED-5: trace length ~150mm + 6 vias → 40MHz unreliable)
- SNES ROMs: 256KB to 6MB typical (HiROM/LoROM)
- FAT32 filesystem for easy ROM management via PC
- 3.3V logic: no level shifter needed (module has built-in)
- SPI pins: GPIO38/39 (CLK/CS) + GPIO43/44 (MISO/MOSI) — split across WROOM-1 left and right
- v2 TODO (R4-MED-1): add local 100nF bypass within 5 mm of U6 VCC

MCU – ESP32–S3–WROOM–1 N16R8

Dual–core LX7 @ 240MHz, 16MB Flash, 8MB PSRAM

GPIO ASSIGNMENT TABLE

Display (8080 parallel):
GPIO4–11 = D0–D7 (8–bit data bus)
GPIO12=CS GPIO13=RST GPIO14=DC
GPIO46=WR (RD/BL tied +3V3 on FPC)

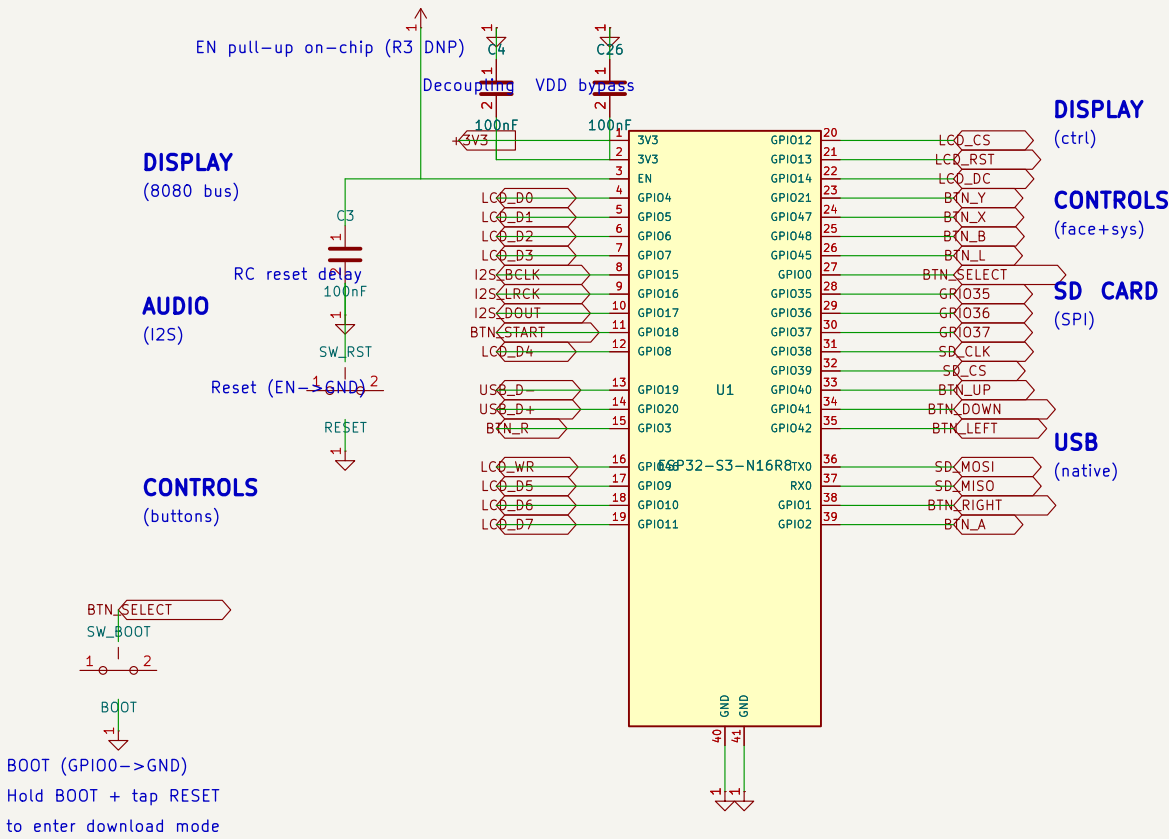
Audio (I2S):
GPIO15=BCLK GPIO16=LRCK GPIO17=DOUT

SD Card (SPI):
GPIO44=MOSI GPIO43=MISO GPIO38=CLK GPIO39=CS

Controls (active low, 10k pull–up):
D–pad: GPIO40=UP 41=DOWN 42=LEFT 1=RIGHT
Face: GPIO2=A 48=B 47=X 21=Y
Sys: GPIO18=START 0=SELECT 45=L 3=R

USB (native, firmware flash + debug):
GPIO19=D– GPIO20=D+

Reserved (do not use, module–internal):
GPIO26–32 = Octal Flash (N16)
GPIO33–37 = Octal PSRAM (R8)

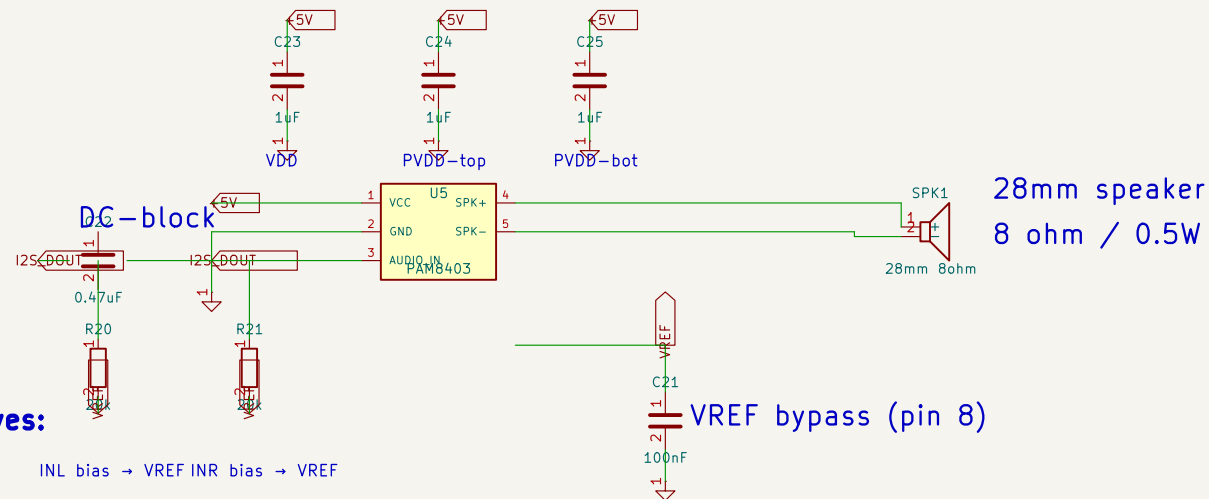


AUDIO OUTPUT

I2S DAC → PAM8403 Class-D Amplifier → 28mm 8ohm Speaker

I2S Bus (PDM TX mode – DOUT only):

I2S_BCLK	GPIO15 – Bit Clock (UNUSED – PDM mode, v2 free)
I2S_LRCK	GPIO16 – L/R Word Select (UNUSED – PDM mode, v2 free)
I2S_DOUT	GPIO17 – PDM sigma-delta out @ 32kHz → PAM8403



Design Notes:

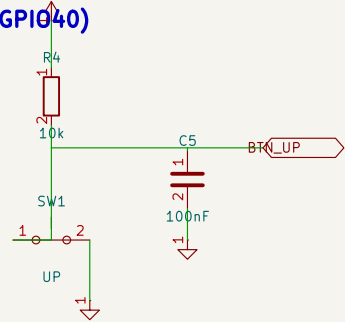
- PAM8403: filterless Class-D stereo amplifier (3W per channel max)
- Only one channel used for mono audio
- No external DAC needed: ESP32-S3 has built-in I2S with DMA
- Powered from +5V rail for max headroom (3.3V limits volume)
- DMA-driven audio streaming for low CPU overhead

CONTROLS – 12 Tact Switches

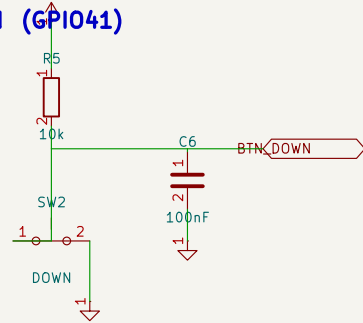
Active-low with 10k pull-up + 100nF debounce per button

D-PAD

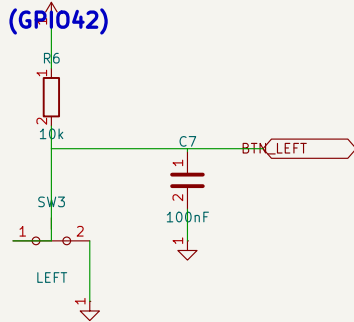
UP (GPIO40)



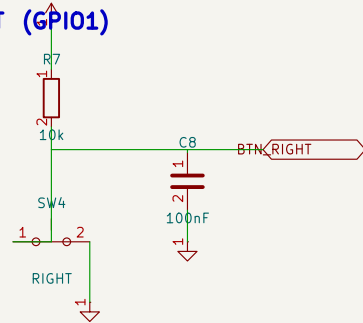
DOWN (GPIO41)



LEFT (GPIO42)

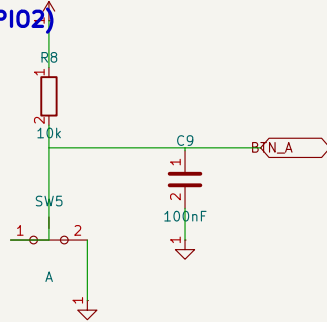


RIGHT (GPIO1)

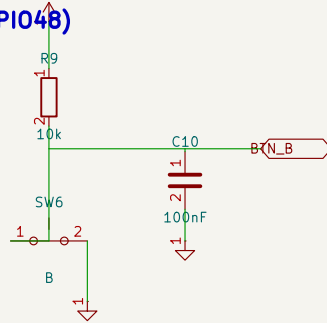


FACE BUTTONS (ABXY)

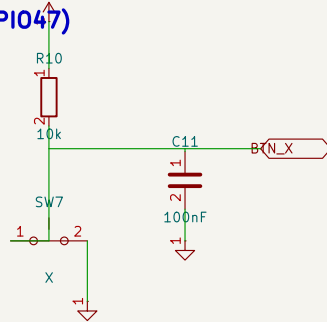
A (GPIO2)



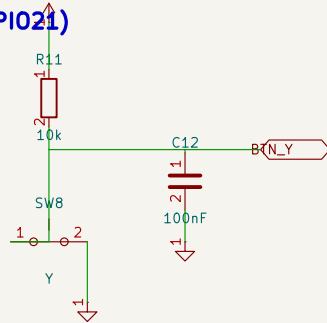
B (GPIO48)



X (GPIO47)

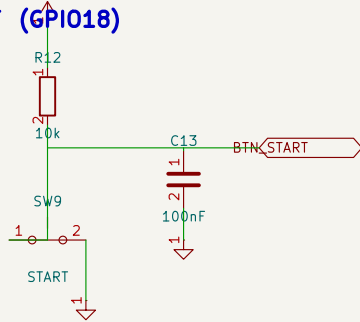


Y (GPIO21)

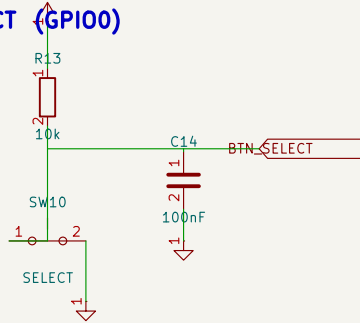


SYSTEM + SHOULDER

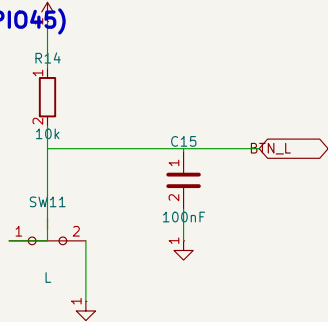
START (GPIO18)



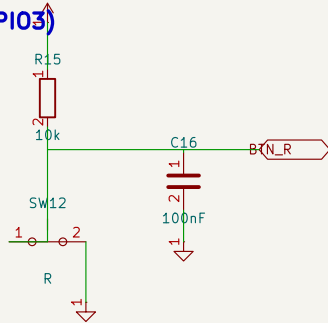
SELECT (GPIO00)



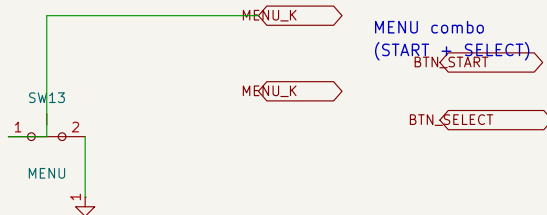
L (GPIO45)



R (GPIO3)



MENU BUTTON



BUTTON CIRCUIT (repeated 12x):